

4.3 Proof of Upper Bound of Theorem 2

We aim to prove Theorem 2 by induction n . Clearly it is enough to show Theorem 2 for all induced subgraphs of Λ_U . The base cases are when $n \leq 7$, which are easily verified. For the inductive step, suppose $n \geq 8$ and $\Lambda_U[S]$ is an n -vertex subgraph of Λ_U with e edges. Additionally, suppose $\Lambda_U[S]$ has the maximum number of edges out of all n -vertex subgraphs of Λ_U . The inductive hypothesis is that all n' -vertex subgraphs of Λ_U , where $3 \leq n' < n$, have at most $e(n')$ edges.

Claim 15. *We may assume $\Lambda_U[S]$ is 2-connected.*

Proof. $\Lambda_U[S]$ must be connected, otherwise we could translate a connected component of $\Lambda_U[S]$ to form additional edges contradicting the maximality of $\Lambda_U[S]$. Suppose $\Lambda_U[S]$ had a cut vertex v . When we remove v from $\Lambda_U[S]$ we create two connected components $G_1 = \Lambda_U[S_1]$ and $G_2 = \Lambda_U[S_2]$ with n_1 and n_2 vertices, and e_1 and e_2 edges.

Case 1: $n_1 < 4$ or $n_2 < 4$. Without loss of generality suppose $n_1 \leq 3$. If $n_1 = 1$, there is 1 edge removed when deleting S_1 from $\Lambda_U[S]$. Applying the inductive hypothesis to the remaining graph we obtain $e \leq e(n-1) + 1 \leq e(n)$. So supposing that $n_1 \in \{2, 3\}$, then there are at most 6 edges removed when deleting G_1 from Λ_U . Then the inductive hypothesis implies

$$e \leq 6(n - n_1) - 4\sqrt{6(n - n_1) - 6} + 6 \leq 6n - \sqrt{96n - 63} \text{ for } n_1 = 2, 3 \text{ if } n \geq 6.$$

Case 2: $n_1, n_2 \geq 4$. Applying the inductive hypothesis to G_1 and G_2 we obtain

$$\begin{aligned} e &\leq 6n_1 - 4\sqrt{6n_1 - 6} + 6n_2 - 4\sqrt{6n_2 - 6} + \deg(v) \\ &\leq 6n - 6 - 4\sqrt{6(n - 5) - 6} - 4\sqrt{18} + \deg(v). \end{aligned}$$

We can bound the degree of v by noticing the neighbourhood of v must be disconnected. It is easy to see through Menger's theorem that the neighbourhood of v in Λ_U is 4-connected, which implies that the degree of v in $\Lambda_U[S]$ is at most 8, hence

$$e \leq 6n - 6 - 4\sqrt{6(n - 5) - 6} - 4\sqrt{24} + 8 \leq 6n - \sqrt{96n - 63} \text{ for } n \geq 7. \quad \square$$

Claim 16. *We may assume there is no line parallel to an element in U intersecting Λ but disjoint from S , and with vertices from S on either side of the line.*

Proof. Suppose there is such a line L . Since $\Lambda_U[S]$ is 2-connected there must be at least two edges of $\Lambda_U[S]$ that cross L . If L is parallel to a short edge (say parallel to g_1), the only edges of $\Lambda_U[S]$ that can cross L will be long edges perpendicular to L (in direction $g_1 - 2g_2$). We shift a set of vertices on one side of the line towards L along one of the directions in U . This is depicted in Figure 11, where the part of S above L is shifted by $-g_2$. Notice that each